



Smart Walker Control System for User Fall Detection and Postural Stability Adjustment Through Planned Smart Walker Motion

A project report presented to the faculty of the
Department of Mechanical Engineering
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In partial fulfillment of the requirement for the Degree
Master of Science

by

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Revised: December 17, 2025

Smart Walker Control System for User Fall Detection and Postural Stability Adjustment Through Planned Smart Walker Motion

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Abstract—Smart walkers enhance mobility assistance for users by utilizing motorized wheels and control systems, while providing the postural support of traditional mobility walkers. Smart walkers can misinterpret a user falling as aggressive input, causing unintentional smart walker movement. This project seeks to implement a fall detection and countermeasure path planning system for the San Jose State University Assistive Robotic and Medical Systems Smart Walker. A fall detection system was developed, user force input to experimentally derived threshold values for force torque deviation and rate of change. The fall detection system was developed to detect Forward, Left, Right, and Backward direction falls. Of 20 fall detection trials, 16 falls were correctly detected, resulting in 80% detection accuracy. A fall countermeasure motion plan was designed, commanding an angular displacement of the heading of the Smart Walker to align the heading with the direction of the fall force vector, providing user postural adjustment through a simultaneous linear displacement along the heading axis. The countermeasure motions were tested and evaluated by how effectively the user acceleration was brought to the lowest value and stabilized. The final configuration of the fall countermeasure motion improved WTI to below 1.0 within 0.78-1.08s for each fall direction.

I. INTRODUCTION

Mobility walkers are essential tools to provide mobility assistance for those experiencing inabilities and deficiencies in their gait. While this includes those experiencing lower-body injuries or physical disabilities that affect walking, mobility walkers are aimed primarily at the elderly. The World Health Organization (WHO) estimates 684,000 fatalities are due to falls, with “adults older than 60 years of age suffer the greatest number of fatal falls” [1]. Furthermore, the WHO estimates that an additional 37.3 million falls occur annually and are severe enough to require medical attention, potentially leading to injuries that affect the quality of life [1]. A common tool for fall prevention in the elderly is a walker. A traditional walker provides support to keep the upper body upright with arm supports, as well as provides lateral stability with their wide wheelbase. This is essential for the elderly losing physical functions such as muscle strength, balance, proprioception, and muscular endurance. In addition to the physical aid a walker provides, walkers also give the elderly a sense of independence by providing them elderly with the means to navigate their home and the outside world without the assistance of another person. However, even with traditional walkers providing postural support, users are still at risk of falling during use. In a study conducted on walker design deficits by Nickerson et

al. [2], 27 fall incidents were observed between two elderly long-term care facilities involving rollator-type walker users. Of these 27 fall incidents, 51.85% were during “Turning”, “Walking Forward”, and “Standing” situations, or normal expected walker usage. There are multiple autonomous mobility walkers in development such as the Movclinic [3], a robotic walker with an active-passive hybrid actuator [4], a scaleable dynamic obstacle avoidance smart walker [5], a mobility assistive device with a parallel manipulator [6], the ReRobo Walker [7], and Nessie [8]. Each has a similar purpose of assisting the user using the walker, but with a different problem focus. A smart walker prototype developed in Malaysia by Asante et al. focused on developing a smart walker with obstacle avoidance by adding minimal modular systems for object detection, and autonomous and manual operation of electric drive motors on standard rollator walkers for simpler replication. Its key feature was the ability to autonomously return to the user using obstacle avoidance and facial recognition after a voice command, while keeping the overall design simple and low-cost for “a blueprint for easy replication and incremental development, especially considering that the larger (hardware) components utilized may be significantly downgraded for more affordable alternatives without significantly affecting the prototype’s quality or behaviour.” [5]. The Movclinic is a motorized walker that also includes “an obstacle detection system which is based on the ultrasound principle, generating an audible alarm when detecting them” [3]. The preliminary focus of the Movclinic was assisting the walking gait of the user and utilizing an obstacle alarm system, which would require path correction by the user. The robotic walker utilizes a novel hybrid actuator comprising a rotary magnetorheological brake and DC motor, aiding in uphill and downhill travel [4]. Traditional walkers work best on flat surfaces, as slopes provide higher user balance and leg strength to maintain stability and control of speed. The mobility assistive device with the parallel manipulator focuses on having a motorized walker base with an assisting parallel manipulator that assists the user both rising into and lowering from a standing position [6]. The team developing the walker focused on giving more assistance in the standing position, a function not provided by traditional walkers found to be challenging by a percentage of walker users. The ReRobo Walker is the most similar in desired functions of an assistive walker currently. It utilizes a motorized base that also has a sensor array including LiDAR and force sensors. The walker can provide assisted autonomous travel to preset locations using motion planning and active obstacle detection, as well as fall

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detection with the force sensors [7]. The Nessie smart walker focused primarily on allowing the walker to carry heavier loads over uneven outdoor terrain, including the user, with a vision system for obstacle avoidance and object recognition for visually impaired [8]. Currently, the autonomous walker field converges on a motorized base that assists the user in walking. Their differences lie primarily in a focus on a different user need, such as the sit-to-standing motion [6], walking on a slope [4], or carrying heavy loads for the user []. Some of the walkers implement fall detection methods such as the ReRobo [7], but none of these smart walker designs implement a fall response system to counteract user falling. A major usage category for smart walkers is the use in assisting rehabilitation for those with gait imbalance. The AGoRA smart walker was developed with user gait estimation through analysis of forces and torques applied to the user handle, and remote control systems for therapy managers to control the AGoRA parameters and control strategies. Gait estimation data is saved and able to be accessed by therapy managers for further analysis for rehabilitation planning [9]. A smart walker developed by Chalaki et al. performed a gait analysis of rehabilitation patients by analyzing the applied torques to the user handle. The main focus was identifying gait asymmetry, “allowing for the extraction of spatiotemporal variables such as step length, step time, and stance or swing phase duration” [10]. A smart walker developed by Ji et al. was designed to assist rehabilitation patients’ walking training by utilizing force sensors connected to sliders that make contact with the user’s pelvis. This allows for the smart walker to drive the electric motors on the corresponding side of the pelvis moving forward, as well as adjust the motor speed by how much force is applied by the pelvis [11]. In all cases, the smart walkers were developed to have novel control methods to detect patient intent while walking and utilize control systems to keep the walker in a position of postural support and provide light locomotive assistance. Currently, there are various means of fall detection, divided by the sensors used. These sensors include accelerometers, frequency shifts in radio waves, camera image processing, and tactile force sensors. Accelerometers are the most common sensor used in fall detection. Accelerometers can be used in wearable devices, as the accelerometers themselves have a small form factor. This can lead to simple wearable devices such as the belts used in the study by Sriraam et al. [12]. While good at detecting changes in velocity, the system required defining a threshold of magnitude of acceleration that was considered the lower limit to define a false alarm zone. With use in a walker, there would need to be a sensor on the walker and the user to confirm falling before path correction but would need to do that with a smaller detected acceleration to give the walker the time to change the motion path to correct the fall. However, the accelerometer can be placed in a different location with a smaller tolerance, such as on the foot, akin to the Intelligent Foot Orthosis [13]. This device focuses on detecting the angle change of plantar dorsiflexion corresponding to a side fall. While a more

specific mode of falling, this device shows that the placement of the accelerometer would influence the acceleration data reading speed. Radio wave frequency shift detection using an RFID passive tag-based sensor for fall detection was used by Huang et al. in their novel approach to fall detection [14]. This sensor system requires a wearable tag and would be reliant on a difference in motion with the RFID tag and a radio wave sensor on the walker itself. While low cost, the accuracy and need for the sensed falling being the user away from the walker make the RFID tag-based sensing in the motion path correction system. Fall detection by camera involves processing information from an image based on key features and changes in the definition or orientation of the feature. A fall detection system was developed that utilized a cell phone as a wearable sensor, utilizing image processing from the camera’s video recording, judging a fall by changes in orientation of detected edges of objects in the image [15]. This method can have moderate accuracy, but relies on good image recording and defined edges, making it more suitable for smaller indoor spaces where walls can be utilized as edge references. A tactile force sensor would provide a more accurate sensor at the cost of a more complicated system, as seen in the fall detection device developed by Li et al. [16]. This device utilizes a series of small force sensors that detect forces along the bottom of the foot. This system requires specific force maps that determine normal force shifts along the foot during various positions during the user’s average walking gait, and then determine falling by analyzing the out-of-expected force patterns. This system is accurate and sensitive, but may have complexities when gait patterns may have large variances while using a walker. Some of the developing smart walkers attempt to integrate fall detection. Some are based on singular sensors. The StrideSafe is designed to be a small attached device to a walker that utilizes an IMU-sensor to detect 6-DoF acceleration of the walker for fall detection [17]. The StrideSafe uses machine learning to develop microcontroller recognition of fall accelerations after training. While simple to implement, fall detection for single-sensor systems relies on accurate models for fall scenarios achieved through experimental development or machine learning. False detections can be common in single-sensor setups. Most smart walkers implementing fall detection utilize multi-modal methods of fall detection. A smart walker by Cavaglià et al. combined sensor data from an IMU on the back of the user and a laser range finder to combine acceleration data with user leg position and stride length for fall detection [18]. This does require the user to wear a sensor, which can be uncomfortable, and the user can also forget to wear the sensor, eliminating a fall detection mode. The ReRobo smart walker combined data from the force-torque sensor and LiDAR for fall detection [7]. As the sensors are integrated into the walker, fall detection is done without the user wearing a sensor. The fall detection system in this case will have to create an estimate of the user’s acceleration or body position to model if the user is in a fall state. In multimodal fall detection, sensor fusion is performed to combine sensor data from homogeneous or heterogeneous

types of sensors to determine if a fall incident has occurred. A high-reliability fall detection wearable was developed by Pierleoni et al., combining the data from the 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer by quaternion representation of orientation [19]. Quaternion representation creates an orientation estimate by combining the estimated orientation from each sensor with the estimated rate of change of orientation from each sensor, giving a weight to either orientation calculation. This method works for sensor fusion of sensors with similar estimates, in this case, orientation. This method of sensor fusion is categorized as feature-level fusion. The wavelet transform is another feature level of sensor fusion. Wavelet transforms sensor signals by centering the sensor signal where a rapid deviation from transient behavior is observed. In the studies of [20] and [21], acceleration spikes from multiple accelerometers are the basis of the wavelets, and are compared to a mother wavelet. The mother wavelet is an established wavelet of the acceleration during a fall. The wavelets are used to establish fall detection by a wavelet of a sensor exceeding the threshold established by the mother wavelet. The acceleration wavelets detected are a feature of the sensor fusion, establishing fall detection by comparing if sufficient acceleration is detected by any of the sensors. The next level above the feature-level sensor fusion is decision-level sensor fusion. Decision-level sensor fusion takes features detected by the individual sensors, determines an environment from the combined features, and takes action corresponding to whether the environment necessitates action. In a study by Xefteris et al., the benefits of sensor fusion of unrelated sensors were discussed for fall detection [22]. The study looked into the combination of wearable sensors, such as IMUs, and non-wearable sensors such as ambient radar, and image processing from cameras. The study detected falls by determining fall parameters for each system of sensors and detecting falls by confirming if fall conditions were met by multiple systems. The benefit of combining multiple heterogeneous sensor systems was maximizing particular features of each sensor system, such as “triggering can be fast and lightweight” for the wearable IMUs, and having the environmental sensors constantly powered for long-term fall detection [23]. Each sensor fusion algorithm has its strengths and weaknesses. A study done by Tsinganos et al. compared 4 sensor fusion algorithms of accelerometer, gyroscope, and magnetometer sensors for fall detection, focusing on performance based on fall detection sensitivity, specificity, and precision. The best-performing algorithm performed feature-level fusion, calculating acceleration, angle variation, angular velocity, and angular acceleration from each sensor. These feature values were averaged and compared to establish thresholds to determine falls. Smart walker technology development has the intent of using motorized wheels to assist in the walking of the user, with the different walkers having different user needs focus. Smart walkers that feature fall detection currently do not show features that maintain user postural stability in detected falls, other than stopping the motors as seen in the ReRobo. The challenges of implement-

ing a fall detection and postural stability maintenance control system for a smart walker will be accurate fall detection and impactful walker motion to catch the user before the fall leads to injury. Current fall detection technology shows there is often the need to create specific parameters for fall conditions that cannot be mistaken for incorrect user control. The fall detection system will require thorough consideration of different fall scenarios and sufficient testing to ensure the parameters of a fall are accurately defined for the sensors. The path planning for a smart walker to maintain user postural stability will need to utilize motion path planning with considerations for the limitations of being driven by rear motors and limited direction change capabilities. The planned path will need to sufficiently position the walker in a position of support and be physically possible for the walker while avoiding obstacles to effectively catch the user.

The objective of this proposed project is to implement user falling detection and path planning for the falling motion on the ARMS Lab Smart Walker Version 2 to maintain user postural stability. The response paths are to be primarily through utilizing specific motor movement protocols in response to detected specified fall situations. Currently, the ARMS Lab Smart Walker does not have any safety measures for the situation where the user falls. In fall scenarios, the Smart Walker may read the force applied to the handles during falls as input for walker velocity control. The Smart Walker will then move inappropriately, leading to worse injuries to the user and possible damage to the Smart Walker. Fall detection and path planning responses enhance the user safety of the ARMS Lab Smart Walker, increasing the capabilities and trust the user has in the Smart Walker. Furthermore, no comparable smart walkers in development have shown developed control systems to have the walker move and catch the user falling in response to a detected fall, as planned for this project. This project will bring the ARMS Lab Smart Walker to the forefront of smart walker technology, adding a fall detection system without the need for user-worn equipment, and a novel approach to counteracting the user center of mass acceleration during falls.

II. METHODOLOGY

The Assistive Robotics and Medical Systems (ARMS) Lab of San Jose State University, directed by Dr. Mojtaba Sharifi, seeks to further enhance the assistance a mobility walker provides by developing an autonomous motorized smart walker. The smart walker is labeled as Smart Walker by ARMS Lab and can be seen below in Figure 1.

The Smart Walker utilizes motorized wheels, LiDAR, cameras, and a 6-axis force sensor, along with machine learning algorithms for environment mapping for autonomous course correction for obstacle avoidance. The user handle is designed to support the user’s forearms while grasping the handle, providing postural stability while walking. The user handle is connected to the frame by the ATI Mini58 6-axis force sensor, providing a means to control the Smart Walker path by force and torque application to the handle.

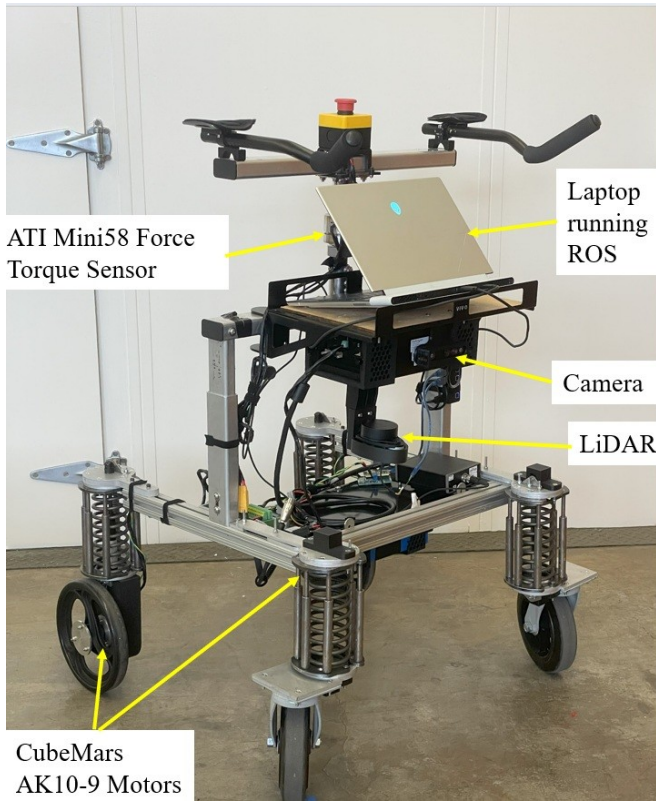


Fig. 1: ARMS Lab Smart Walker Version 2. It utilizes two CubeMars AK10-9 brushless DC motors on the rear wheels to drive the walker, and a 6-axis ATI Mini58 force torque sensor attached to the user handle for command inputs. The Smart Walker also has a LiDAR and a camera for autonomous control.

This project focuses primarily on the development and integration of the sensor hardware and control software systems for fall detection and path planning to maintain postural stability in fall scenarios on the existing ARMS Lab Smart Walker system. Building upon the existing Smart Walker necessitates developing and integrating the system in the Robot Operating System (ROS) coding framework currently used for the Smart Walker, as well as utilizing the existing 6-axis force sensor and LiDAR hardware for fall detection. The fall detection system will categorize the fall into a generalized fall mode categorized by primary force and torque direction applied to the user's handle. When a fall is detected, the control system calculates the angle and magnitude of the applied force vector in the X-Y plane to determine the direction and severity of the fall. In response to the applied force vector to the handle by the fall, the Smart Walker control system sends commands to the motors of the rear wheels to create angular displacement to orient the Smart Walker heading to the direction of the force vector, and command linear translation to provide the counteracting force to the user to improve postural stability.

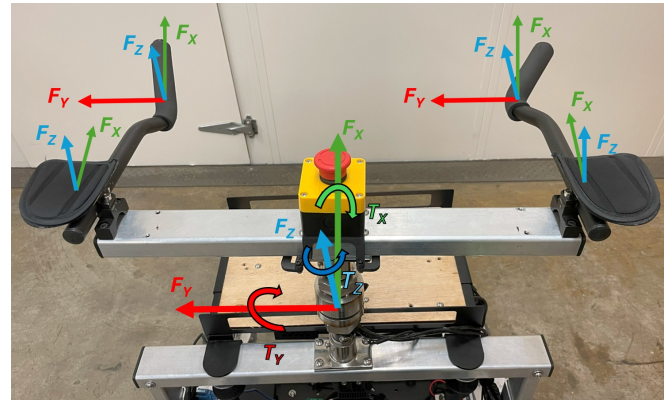


Fig. 2: Forces applied at the end of the handle by the user's hands, and at the elbow pads where the elbows rest. During falls, no Y-axis forces are applied by the elbows, only applying torques to the support column. All forces are combined into a force and torque vector applied to the support column, registered by the integrated ATI Mini58 force torque sensor.

A. Fall Detection

Fall detection is done primarily by comparing force torque input to the user handle to threshold force torque deviation and rate of change values, detecting occurring falls, and categorizing them by the primary axis acted on. The user handle translates forces applied by the user's hands at the end of the handles, as well as by their elbows, as contact points on the user handle structure. These forces are read by the ATI Mini58 force torque sensor located in the connection point of the user handle and the Smart Walker structure. Due to the design of the handle, forces applied by the user can be combined into a force vector, as well as torques applied to the Mini58 force torque sensor. Figure 2 shows the forces applied at the contact points of the user handle and how the Mini58 force torque sensor interprets the input.

B. Fall Scenario Force Torque Parameter Identification

The main focus of the initial work of the project was deriving the threshold values for the forces and torques applied to the user's handle during falls. These threshold values would be the basis for detecting if the user was falling. Force and torque data were recorded during mock fall scenarios. Average values for the forces and torques were derived from trial data. Table 1 shows the average force and torque values applied to the user handle during the "Forward", "Left", "Right", and "Backward" fall scenarios. Table 2 lists the force and torque deviation from normal input on the user handle during forward walking during the "Forward", "Left", "Right", and "Backward" fall scenarios.

C. Creating the Fall Detection and Countermeasure System in the ROS Framework

The coding framework of the Smart Walker is ROS. The fall detection and countermeasure system builds on the current Smart Walker control system by adding a fall

TABLE I: Average Force and Torque Inputted During Fall Scenarios

Scenario	X Force (N)	Y Force (N)	Z Force (N)	X Torque (Nm)	Y Torque (Nm)	Z Torque (Nm)
Normal Walking	53.461	10.255	-236.440	5.568	29.777	1.510
Forward Fall	98.479	10.211	-377.400	-7.023	28.548	-0.600
Left Fall	54.231	20.705	-207.600	-66.743	4.184	-7.600
Right Fall	50.343	-34.414	-578.600	83.411	2.556	15.800
Backward Fall	-80.721	12.296	-208.200	7.130	-51.621	0.200

TABLE II: Force and Torque Deviation from Normal Walking Input During Fall Scenarios

Scenario	X Deviation (N)	Y Deviation (N)	Z Deviation (N)	Tx Deviation (Nm)	Ty Deviation (Nm)	Tz Deviation (Nm)
Forward Fall	45.018	-0.044	-140.960	-12.591	-1.229	-2.110
Left Fall	0.770	10.450	28.840	-72.311	-25.593	-9.110
Right Fall	-3.119	-44.669	-342.160	77.843	-27.221	14.290
Backward Fall	-134.182	2.042	28.240	1.562	-81.398	-1.310

detection component to the Velocity publisher, which is the node that takes force torque readings and calculates a desired linear and angular velocity based on the force torque input. By putting the fall detection in this node, the fall detection can occur before the commanded velocities are published. If a fall is detected, the direction and magnitude of the force vector in the XY plane are calculated and sent to an action server node to perform the countermeasure motion. Doing so prevents the Smart Walker from sending velocity commands based on the input on the handle.

The fall detection system compares the deviation of each force and torque axis between sensor readings. The rate of change calculation is also performed by dividing the force torque deviation vector by the time period between readings. To improve the performance of the fall detection system, the fall detection node compared the force torque deviation every 0.1s and calculated the rate of change of the force torque input every 0.05s. By checking the rate of change more frequently, the fall detector has more sensitivity to falls where the force applied is under thresholds, looking for a significant impulse indicative of a user falling.

The fall detector compares the deviation and rate of change of the forces and torque values to the thresholds identified experimentally through mock fall trials. If the threshold is met or exceeded for a particular fall scenario, the fall detector node would change a fall state boolean to true, triggering the fall countermeasure action. The fall state boolean disables the force torque input to the velocity calculator, preventing higher motor velocity commands due to increased input on the user handle. Figure 3 combines a higher-level representation of the Smart Walker control system with the implementation of the fall detection and countermeasure action server.

D. Smart Walker Kinematics

The Smart Walker Version 2 is driven by two electric brushless motors, the Cubemars AK10-9 motors. These motors drive the rear wheels of the walker that are fixed to the walker's frame. The Smart Walker Version 2's movement is therefore derived from the forward kinematics of a differential wheeled robot, where steering is accomplished

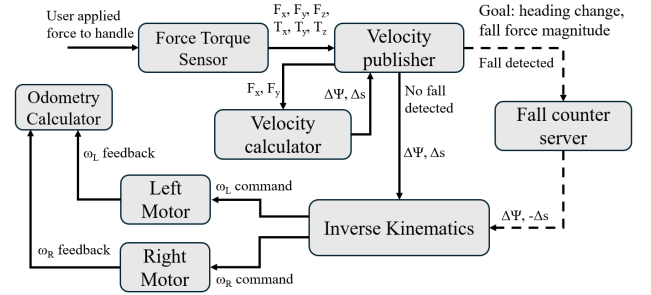


Fig. 3: Flowchart representation of the Smart Walker control system with the implemented fall detection and countermeasure control system. The velocity publisher compares the force torque input to the fall threshold values. If a fall is detected, a ROS action server goal of the fall vector direction and force magnitude is sent to the fall counter action server to calculate a desired $\Delta\psi$ and $-\Delta s$ movement. This bypasses the use of force to derive the desired motor velocity.

by varying the relative rate and direction of rotation of the drive wheels. Figure 4 shows a diagram of the Smart Walker differential drive motion.

The forward kinematic equations for change in angular position ψ and linear displacement Δs due to the relative angular velocities of the drive wheels are in Eqs 1 and 2 below, where $d\theta_r$ and $d\theta_l$ are the angular displacements of the right and left wheel, D is the wheel diameter, and T is the distance between the wheels.

$$\Delta s = \frac{1}{4}D(d\theta_r + d\theta_l) \quad (1)$$

$$\Delta\psi = \frac{1}{2}D\frac{(d\theta_r - d\theta_l)}{T} \quad (2)$$

The inverse kinematics of the motion of the Smart Walker Version 2 is utilized in the control system trajectory input. In normal operation, the forces and torques are used to calculate a desired angular velocity and linear velocity depending on the input forces in the X and Y directions. The angular velocity input and linear velocity input are sent to a node that combines the desired inputs and derives a left and right wheel velocity utilizing inverse kinematics for differential

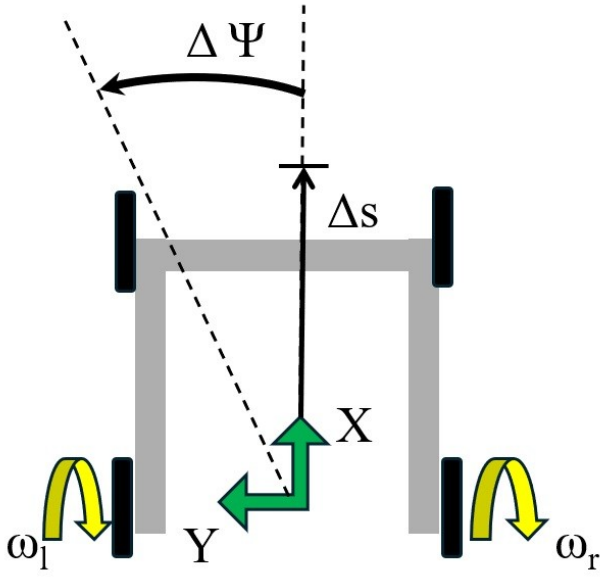


Fig. 4: Smart Walker differential drive robot diagram. Linear translation along heading direction Δs and angle change in heading direction $\Delta\psi$ are due to difference in rear wheel motor speeds ω_l and ω_r .

drive robots. The inverse kinematics equations can be seen in Eqs. 3 and 4, where D is the diameter of the wheels, and T is the distance between the wheels.

$$\frac{d\theta_r}{dt} = \frac{8}{D} \frac{\Delta s}{dt} + \frac{4T}{D} \frac{d\psi}{dt} \quad (3)$$

$$\frac{d\theta_l}{dt} = \frac{8}{D} \frac{\Delta s}{dt} - \frac{4T}{D} \frac{d\psi}{dt} \quad (4)$$

E. Smart Walker Fall Countermeasure Motion Planning

The direction of the fall was found by finding the angle of the XY force vector from the positive x-axis. As the positive x-axis aligns with the walker's heading, the angle of the force vector is the required angular displacement the Smart Walker needs to rotate to align the heading to the fall direction. The angle was found using the `atan2` Python function, giving an angle from 0 to $+\pi$ if the force vector points to the left of the heading, and 0 to $-\pi$ if the force vector points to the right. The angle of the force is then used to create a desired rotation trajectory by multiplying the angle by a sigmoid function, specifically the logistic growth function, to create an angular displacement path plan by utilizing the force direction as the carrying capacity. By finding the derivative of the logistic growth function, an input signal for the angular velocity of the Smart Walker can be calculated. The angular displacement equation can be seen in Eq. 5, and the angular displacement equation can be seen in Eq. 6. where θ is the direction of the force vector, k is the steepness, t is time, and t_0 is the time to reach half of the carrying capacity.

$$f(t) = \frac{\theta}{1 + e^{-k(t-t_0)}} \quad (5)$$

$$f'(t) = kf(t)(1 - f(t)) \quad (6)$$

By utilizing a sigmoid function and the derivative, a smooth angular velocity input can be generated where smooth acceleration at the beginning and smooth deceleration occur when approaching the desired angular displacement. The steepness k can be used as a gain value to increase the angular velocity, while t_0 is used to tune the response time of the angular motion.

To provide postural stability to the user during a fall, the linear translation Δs in the direction of the heading of the Smart Walker is calculated for the Smart Walker to provide deceleration to the user. The magnitude of the XY force vector is calculated and multiplied by a base linear velocity, increasing the Smart Walker linear velocity relative to the severity of the fall force.

Different countermeasure motions implemented different combinations of angular displacement ψ and linear translation Δs in response to the fall direction and magnitude.

- 1) $+\psi$, only rotating the heading of the Smart Walker to align with the applied force vector direction
- 2) $-\psi$, only rotating the heading of the Smart Walker to the opposite direction of the applied force vector direction
- 3) $+\psi$ and $+\Delta s$, rotating the heading of the Smart Walker to align with the applied force vector direction, and adding linear translation in the direction of the fall
- 4) $+\psi$ and $-\Delta s$, rotating the heading of the Smart Walker to align with the applied force vector direction, and adding linear translation opposite the direction of the fall towards the user

Through testing, the best-performing combination of motions was found and implemented in the final fall countermeasure motion plan.

F. Fall Countermeasure Testing Methodology

To test the success of the fall countermeasure motion plan, each motion plan was tested in the following scenarios.

- 1) Stumbling forward while walking forward
- 2) Stumbling to the left while walking forward
- 3) Stumbling to the right while walking forward
- 4) Falling backwards while walking backwards

During each fall scenario, the user's acceleration in the X, Y, and Z directions was tracked with an accelerometer attached to the chest of the user, and the Walker Tipping Index (WTI) was calculated and tracked. The test setup can be seen in Figure 5.

The user's accelerometer data shows how well the user was decelerated during the falls. WTI is a measurement of stability for a walking frame based on the forces applied to a rollator walker, indicating if tipping is likely. The equations for WTI for the anterior, medial left, medial right, and posterior directions can be seen in Eqs. 7, 8, 9, and 10, respectively.

$$WTI_a = \frac{(F_{Yl} + F_{Yr}) * H}{(F_{Zl} + F_{Zr} + W) * R_f} \quad (7)$$



Fig. 5: User performing Right Fall scenario test. The accelerometer tracking user acceleration is located in the worn chest harness. Accelerometer data is not used in fall detection.

$$WTI_{ml} = \frac{(F_{Xl} - F_{Xr}) * H}{(F_{Zr} + W/2) * R_H} \text{ or } 0 \quad (8)$$

$$WTI_{mr} = \frac{(F_{Xl} - F_{Xr}) * H}{(F_{Zl} + W/2) * R_H} \text{ or } 0 \quad (9)$$

$$WTI_p = \frac{(F_{Yl} + F_{Yr}) * H}{(F_{Zl} + F_{Zr} + W) * R_b} \quad (10)$$

A WTI value above 1.0 indicates tipping in that direction is likely. The user's acceleration data and the WTI values during each fall scenario were compared for each countermeasure motion plan to determine which motion plan was the most successful and to be utilized in the final countermeasure motion plan.

III. RESULTS AND DISCUSSION

A. Fall Scenario Threshold Forces and Torques

Threshold values of the forces and torques applied to the user handle that define the fall scenarios were found after experimental mock falls to gather data on the forces and torques applied to the user handle during falls. Figure 6

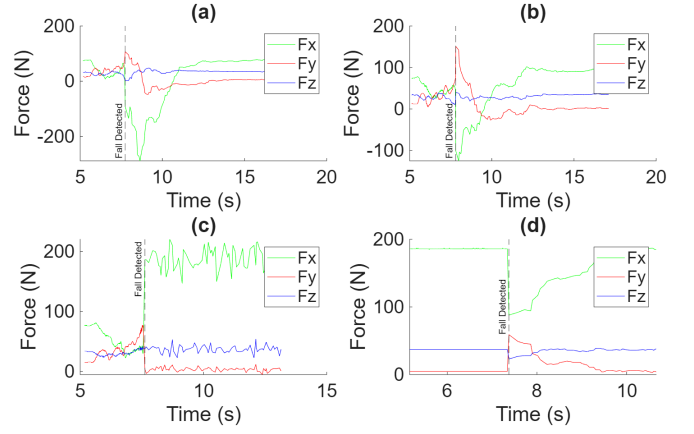


Fig. 6: Forces applied during forward (a), left (b), right (c), and backward (d) fall scenarios and during the fall countermeasure motions for each scenario. Each plot indicates when the fall was detected with a vertical line.

TABLE III: Threshold Values for Force and Torque Deviation and Rate of Change for Fall Scenario Classification

Fall Scenario	Force Deviation (N)	Force Rate of Change (N/s)	Primary Axis	Torque Deviation (Nm)	Torque Rate of Change (Nm/s)	Primary Axis (about)
Forward	70	1200	X	40	700	Y
Left	50	1000	Y	-60	-1000	X
Right	-50	-1000	Y	80	1200	X
Backward	-70	-1500	X	-50	-1000	Y

shows the force input during "Forward", "Left", "Right", and "Backward" fall scenarios.

As opposed to looking at the absolute force and torque thresholds, the fall scenarios were defined by deviation from the prior force-torque reading. This allows for different users who may apply different forces and torques to the handle due to differences in body size or postural positions during normal operations. Furthermore, due to the fall detection system comparing force-torque readings 0.2s apart, the deviation threshold is more aligned with the expected impulses to the forces and torques during falls. The specific fall scenarios have a primary axis that the forces and torques will show the greatest increase during the fall scenario. The fall detection thresholds reflect this. Table 3 summarizes the threshold force torque deviations and rate of changes for the "Forward", "Left", "Right", and "Backward" fall scenarios. In addition to deviation threshold values, the fall detection system also defines falls by rate of change threshold values. These rate of change threshold values allow for falls that do not meet the deviation threshold to be detected due to a large increase in the forces and torques in the detection interval. This is intended to detect less severe falls.

There are some asymmetries between the threshold values of expected symmetrical values, such as the Left/Right falls. This is due to the design of the Smart Walker. There is more play in the user handle connector in the right direction,

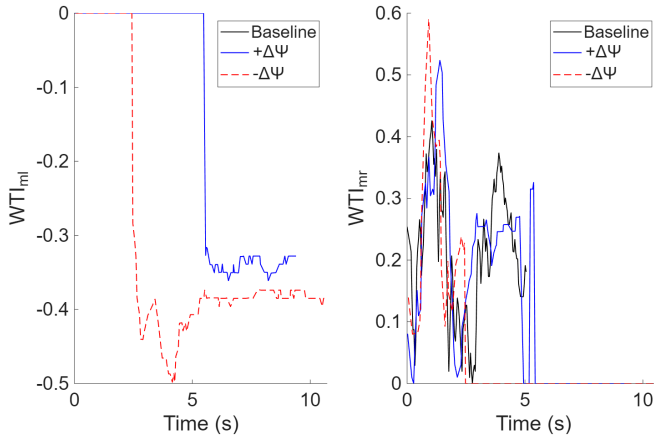


Fig. 7: Comparing WTI_{ml} during Left Fall Scenario (left) and WTI_{mr} during Right Fall Scenario (right) for $+\psi$ and $-\psi$ fall response motion. Utilizing a $+\psi$ motion kept WTI lower during the fall scenario compared to $-\psi$ motion and free-free-moving walker baseline.

allowing for greater force application in that direction.

B. Fall Detection Accuracy

To test the fall detection system, a mock fall was performed during a forward walk. These falls were in one of the four main fall directions: Forward, Left, Right, and Backward. 5 mock falls were performed in each of the main fall directions, for 20 trials. Of the 20 trials, 16 falls were successfully detected for an accuracy of 80%. Failed detected falls had force torque deviation and rate of change below threshold values. During failed fall detection, forces likely are not fully transferred through the user's upper body through to the handle, in cases where the user partially stopped the fall with a reflexive step.

C. Fall Countermeasure Motion Plan Development

Different combinations of commanded angular displacement and linear translation of the Smart Walker in response to the force vector on the XY plane were tested for their effectiveness in decelerating the user and decreasing the WTI. The first tests were performed without any motor commands, allowing the backwheels to free spin to establish a control data set for the user acceleration and WTI. Afterwards, it was tested to compare the effectiveness of pivoting the Smart Walker heading in the direction of the fall ($+\psi$) to align the Smart Walker in a position to provide support to the upper body, or to pivot the Smart Walker to the inverse direction ($-\psi$) about the heading, to attempt to correct the instability. During testing of pivoting the Smart Walker to the $-\psi$ direction, the backwheels were encountering wheelslip and could not overcome the forces applied to the Smart Walker to pivot to the opposite direction of the fall, reducing effectiveness. The pivot to the $+\psi$ direction was chosen to move to the next set of combinations. Figure 7 compares the effect on WTI_{ml} and WTI_{mr} when implementing the $+\psi$ and $-\psi$ motion to the Right Fall scenario.

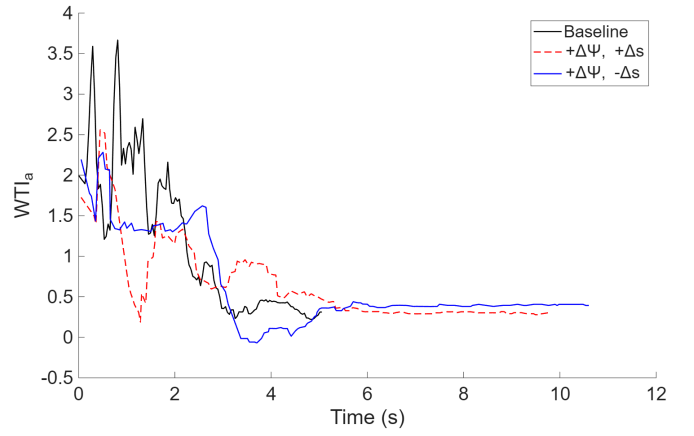


Fig. 8: Comparison of WTI_a during a Forward Fall scenario between $+\Delta s$ motion and $-\Delta s$ motion. $+\Delta s$ provided a sharper decrease in WTI_a , but had a pattern of increase and decrease over time. $-\Delta s$ motion had a more gradual, but consistent decrease in WTI_a .

The next set of combinations tested the effectiveness of improving stability and deceleration by adding a linear translation in the positive direction of the heading ($+\Delta s$) to gradually decelerate the fall, or a linear translation in the negative direction ($-\Delta s$) to push the Smart Walker back into the user to increase postural stability. The most pronounced effect of the direction of the linear translation was observed by observing the WTI_a in the Forward Fall scenario, as seen in Figure 8.

The $+\Delta s$ motion decreased the WTI_a from a peak of 2.5 to close to 0 in 1.2s, a sharper decrease compared to the $-\Delta s$ motion, decreasing from peak WTI_a of 2.28 to 0 in 3.2s. While the decrease in WTI_a was better in $+\Delta s$ motion, it was seen that the additional forward motion of the Smart Walker left the user in an extended unbalanced state. The Smart Walker was not able to stay under the user's upper torso to provide support. This can be seen when comparing the user acceleration in the X-direction. The average user acceleration for the $-\Delta s$ motion was 4.4984 m/s^2 , compared to 4.6999 m/s^2 . While both motion plans achieved bringing the average WTI_a to comparable values of 0.6298 for $+\Delta s$ motion and 0.6480 for $-\Delta s$ motion, the Δs motion provided better user deceleration, and then provided postural adjustment. This was the preferred configuration for the final countermeasure motion plan.

In testing, it was found that in response to a backward stumble fall, the countermeasure motion plan was causing the Smart Walker to perform an aggressive pivot to turn around the Smart Walker heading backwards to align with the fall direction or the inverse fall direction. To prevent the aggressive motion, the countermeasure motion was changed so that if the fall direction exceeded $\pm\pi$ radians, the desired angular position was in the opposite direction of the force vector. This is supported by the user's acceleration during the fall countermeasure motions. When comparing the user's acceleration, the combination of $+\psi$ - Δs motion

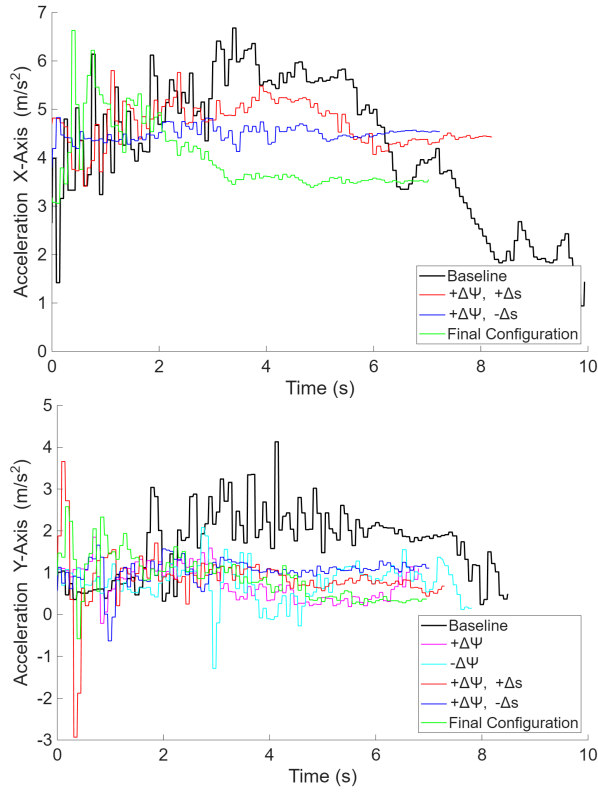


Fig. 9: Accelerations in the X axis for comparing response motion to Forward Fall scenario, and accelerations in the X axis for comparing response motion to Left Fall scenario. The particular axes observed were chosen by the most affected acceleration axes during the specific fall scenario. $+\psi$ and $-\psi$ acceleration data in the X-direction were not included due to the response motion not affecting the Smart Walker motion in the Forward Fall scenario. Utilization of $+\psi$ - Δs motion to counteract the fall showed the most effectiveness in decelerating the user.

reduced the acceleration closer to 0 and stabilized for all fall scenarios. Figure 9 shows the user's acceleration during the fall scenarios.

A final configuration for the countermeasure motion included $+\psi$ - Δs motion. The sigmoid function was tuned to have a steepness k of 8, and t_0 of 0.25s. Figure 10. shows the effect the countermeasure system had on the WTI during Forward, Left, Right, and Backward Fall scenarios.

IV. CONCLUSION

A fall detection and countermeasure motion system to improve user postural stability during fall scenarios for the ARMS Lab Smart Walker Version 2 was designed and implemented. The fall detection system was able to detect user falls with 80% accuracy. The fall countermeasure motion plan implemented a $+\psi$ rotation to the direction of the fall force vector, and added a $-\Delta s$ linear translation. This fall countermeasure motion plan showed the most effectiveness in reducing the user's acceleration during all fall scenarios. This was matched by the reduction and stabilization of the

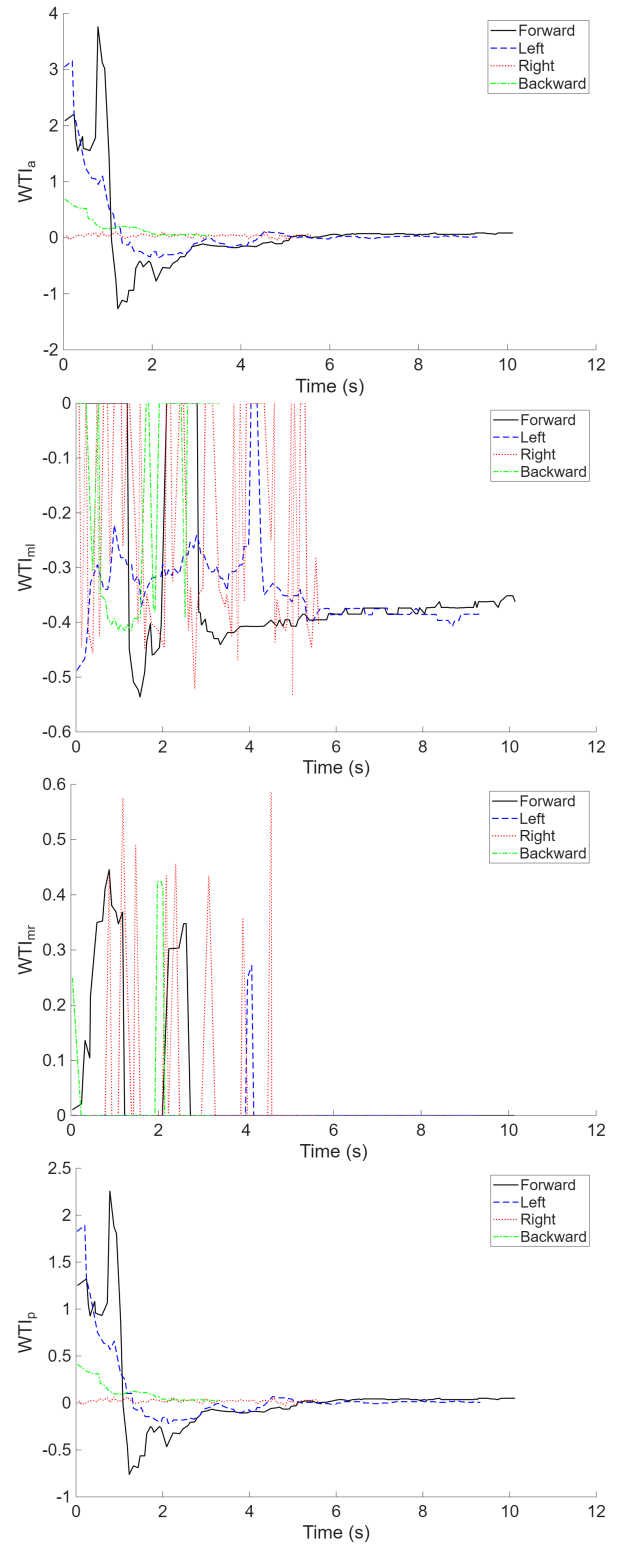


Fig. 10: WTI_a (top left) and WTI_{ml} (top right) WTI_{mr} (bottom left) and WTI_p (bottom right) based on forces applied to the user handle during fall scenarios. The fall countermeasure shows effectiveness in reducing and stabilizing WTI during the forward and backward fall scenarios. The effect on lateral falls is less clear, as there is oscillation in WTI, due to the user resisting the motion of the Smart Walker, providing a higher force to the handle.

WTI_a for forward falls and WTI_p of backward falls. For a Forward Fall Scenario, the final configuration for the fall countermeasure plan reduced peak WTI_a of 3.7622 to 0 in 1.08s, and maintained an average WTI_a of 0.1076 throughout the response motion. WTI_{ml} and WTI_{mr} showed less clear results, due to the rotation of the Walker being resisted by the user's body. This applied more force to the user handle, and had the effect of oscillations in the WTI values.

V. FUTURE CONSIDERATIONS

Improvements to the fall detection system include collecting datasets of force input encountered when intended users fall. The tests were performed by an able-bodied subject attempting to create fall scenarios; therefore, the force-torque deviation and rate of change threshold values may be inappropriate for intended users. The threshold values should be derived from an average user weight, and then be scaled before the Smart Walker is used with respect to the user's weight. This will account for reduced force input by lighter-than-average users, and increased average force input by heavier-than-average users.

The single force torque sensor acting as the connection point of the user handle to the Smart Walker frame has led to inaccuracy issues. The connection of the force torque sensor to the Smart Walker frame has at times become loose, leading to inaccurate force torque readings. In addition, the single connection to the user handle led to discrepancies in certain axes. For example, a discrepancy between the +Y and -Y direction. To improve the design, a two-connection user handle could be implemented, with two force torque sensors acting as the connection to the frame. This would help to eliminate excess movement and deflection of the user handle, increasing force torque reading accuracy.

Fall detection should include additional sensors, such as LiDAR of the user's legs, as a secondary measure of postural instability in situations where the user lets go of the handle during falls. The applied force vector can still be utilized for adjusting the Smart Walker heading and relative position to the user in situations where the user is not fully falling but still requires postural adjustment.

The countermeasure motion plan could be improved by increasing the traction of the rear wheels. During testing, wheelslips occurred due to low traction of the wheels to the lab floor surface, due to the material of the wheels, and shifting of the weight away from the backwheels. The countermeasure motor commands had to consider reducing the speed of the wheels and the direction of the pivot to ensure the motion could be executed effectively. Considerations to rear wheel design include adding a rubber tire to increase traction, or changing the wheel suspension to increase normal force between wheel and floor surface, increasing friction.

The fall countermeasure motion can be improved with a control system integrating feedback from an IMU to better track the angular and linear movement. The feedback to the velocity controller would improve Smart Walker rotation accuracy and account for scenarios such as wheel slip.

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